PHYSICS - STANDARD 9

Chapter: 4

Gravitation

1) The spring stretches when a 100 gram block is suspended from a spring balance held by a finger. Explain the reason for this phenomenon.
(gravity, elasticity, buoyant force, surface tension)
Answer:
gravity

2) The stretching of the spring of a spring balance increases with an increase in the ..... of the block hung on it.
(size, mass, length, weight)
Answer:
mass

3) The force of attraction between two objects is inversely proportional to the ..... between them.
(mass, weight, distance, the square of distance)
Answer:
the square of distance

4) The value of G is 6.67 x 10-11 .....
(N2m2/kg2, N2m2kg2, Nm2/kg2, Nm2/kg)
Answer:
Nm2/kg2

5) If x is the force of attraction with which an object with a mass of 40 kg is attracted by an object with a mass of 80 kg, then the of attraction with which an object with a mass of 40 kg attracts a body of mass 80 kg is .....
(x2
, x , 2x , x2
Answer:
x)

6) The acceleration due to gravity g = ...
(G/MR2, GR2/M, MR2/G, GM/R2)
Answer:
GM/R2

7) The value of 'g' is the maximum ..... of the Earth.
(at the equator, at the centre, in space, in the polar region)
Answer:
in the polar region

8) What is the value of g at the centre of the Earth?
Answer:: zero

9) The force with which an object is attracted by the Earth is its .....
(Mass, volume, density, weight)
Answer:
weight

10) The value of g on the Moon is m/s2
(16.2, 9.8, 1/62, 1.62)
Answer:
1.62

11) The reason why objects moving in a circular path have acceleration even if they have a uniform speed is because .....
(the speed decreases, the direction changes, there is a pulling force towards the centre, they go along a tangent)
Answer:
the direction changes

12) The direction of the acceleration of an object moving in a circular path is .....
(outwards, along the tangent, towards the centre, no acceleration)
Answer:
towards the centre

13) The weight experienced by a freely falling object is .....
Answer:
zero

14) From where do artificial satellites get the required centripetal force?
(Moon, Earth, Rocket, Sun)
Answer:
Earth

15) Mass, the curvature of the curve, and the ..... are the factors that influence the tendency of a vehicle running through the curves to turn around.
Answer:
speed

16) The force of attraction between an object and the earth depends on what factors?
Answer:
the mass of the object and the distance from the centre of the earth

17) What is the reason why the weight of the object at the centre of the earth is zero?
Answer:
Since the value of g at the centre of the earth is zero

18) State the law of universal gravitation.

19) What does each letter in F=GMmR2 represent?

20) What will be the force of attraction experienced when two objects of 40 kg and 50 kg mass are 3 m apart when they are on the moon?
Answer:
m1 = 40 kg
m2 = 50 kg
r = 3 m
F=Gm1m2r2 = 1.48 x 10-8 N

21) The Earth attracts objects with greater force when they have a higher mass. However, a heavy object and a light object dropped from the same height at the same time will fall to the ground at the same time. Why?
Answer:
Because they are equally accelerated by the attraction from the earth or g is independent of the mass of the object.

22) If an object is weighed on a spring balance at the Earth's poles and then transported to the equator and weighed again on the same balance, would there be a change in its measured weight? Justify your answer.
Answer:
There will be a change. Since the value of g is not the same in equatorial and polar regions

23) Why is the value of g different on Earth and on the Moon?
Answer:
g=GM/R2 , because of the difference in the mass(M) and radius(R) of the earth and the moon

24) If a mango and a leaf fall from the same height at the same time then they will not reach the ground at the same time. What could be the reason?
Answer:
Because the leaf experiences more resistance from air

25) What does it mean to be in freefall? What causes you to feel weightless in a freefall?
Answer:
Free fall - Definition
Since the entire weight of the object is used for gravitational acceleration.

26) What is the reason for the acceleration of the object moving in the circular path even if it has uniform speed?
Answer:
As the direction of motion of the object changes.

27) What does it mean by centripetal force? How does the inadequacy of the centripetal force influence the motion of an object?
Answer:
Centripetal force- Definition, if the object in circular motion does not receive sufficient centripetal force, the object will be thrown along the tangent.

28) What are the factors that influence the tendency of vehicles to topple over at curves?
Answer:
Mass, speed, the curvature of the road

29) What is the direction of the centripetal force? Where do artificial satellites orbiting the Earth get centripetal force from?
Answer:
Towards the centre of the earth. From the Earth

30) Calculate the weight of an object with a mass of 20 kg on Earth, the moon and Jupiter in newton(N). (ge – 9.8 m/s2, gm– 1.62 m/s2, gj - 24.5 m/s2)
Answer:
The weight on earth (mg) is 196 N
Weight on the Moon is 32.4 N
Weight on Jupiter is 490 N

31) State the law of universal gravitation. How can this be expressed mathematically? What is the value of the universal gravitational constant?

32) a) When the distance between two objects is halved, the force of attraction .....
(is halved, is doubled, is four times, square times)
b) Two persons standing nearby do not come and touch each other automatically due to the attractive force between them. Why?
Answer:
a) is four times
b) The force of attraction between the person and the earth is very large.

33) Explain the feather and coin experiment.
Answer:
explanation (text book page 79)

34) Skylab, an artificial satellite sent for space exploration, had a mass of approximately 90,000 kg. It stopped working when it was about 435 km above the earth's surface. Due to the mutual force between the satellite and the Earth, the Earth should also have fallen into the Skylab. But the Skylab fell to the earth. Why didn't the Earth fall into the Skylab? Explain the reason.
Answer:
The mass of the earth and thus the force of attraction is very large compared to the skylab, so the acceleration of the skylab towards the earth is very large.

35) Tabulate the differences between mass and weight.
Answer:
differences

36) It is said that gravity is a very small force but influences over wide distances. Establish with examples.

37) a) What is the weight on the earth's surface of an object of 50 kg mass?
b) What is the weight of the object at the centre of the earth?
c) What will be the mass of this object on the surface of the moon?
d) How do you measure mass and weight?
Answer:
a) W = mg = 490.5 N
b) zero
c) 50 kg
d) mass - common balance
weight - spring balance

38) a) What does it mean to be a freefall?
b) What is the reason for experiencing weightlessness in free fall?
c) For an object suspended on a spring balance in the hand of a person sitting on the giant wheel, a difference in weight is shown as the wheel rises and falls. Explain why this occurs.

39) a) What is circular motion?
b) Explain centripetal force by drawing a proper diagram.
c) Write down two examples of circular motion.

40) a) Why do athletes spin and throw a hammer in a hammer-throw competition?
b) In which direction will the hammer be thrown off?
c) What are the factors deciding the distance to which the hammer is thrown off?